

KEYLEARNINGS

AI-Driven Tools for Detecting and Monitoring Mental Health Conditions Through Behaviour Patterns

- AI technique is Natural Language Processing (NLP), which analyses speech and text to identify linguistic and tonal indicators associated with mental health issues
- For instance, studies have revealed that people who are depressed frequently speak more slowly and use harsher words and first-person pronouns
- computer vision enables AI systems to decipher visual clues, including body position, gestures, facial expressions, and micro-expressions, all of which can provide important information about a person's emotional state
- Digital biomarkers for mental health disorders can be found in physical activity levels, sleep cycles, speech tones, social media usage patterns, and typing speed.
- Targeted interventions are made possible by the chatbot's usage of AI algorithms to identify user-specific communication patterns and emotional indications.
- Speech Data: Examined for indicators of disorders like anxiety or depression, such as pauses, tone, and speech pace.
- Text Data: Contains written correspondence, including emails, posts on social media, and transcripts of therapy sessions. NLP algorithms are used to extract linguistic markers such as sentiment and word choice.

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- Physiological Data: Obtained from wearable technology, which monitors parameters including heart rate, sleep habits, and degree of physical activity.
- Use of social media to identify changes in mood, passive data gathering looks at interaction frequency, post sentiment, and engagement patterns
- Supervised Learning ,Unsupervised Learning
- Data privacy is guaranteed by encryption methods and federated learning models. Bias is reduced when models are trained on a variety of datasets.
- Explainable AI (XAI) frameworks increase patient and physician trust by bringing transparency to AI decision-making
- Artificially created data that replicates the statistical characteristics of real-world data, while excluding any personally identifiable information, is known as synthetic data
- genuine mental health data frequently contains sensitive information about people's emotional states, psychological problems, and behavioural patterns, gathering and disseminating it might present ethical and privacy issues.

KEYLEARNINGS

Predictive modeling in MH disorders

- Two prediction techniques, such as fusion models and transfer learning models
- prospective models
- prognostic model is a mathematical system or algorithm designed to estimate the probability or risk of a specific future event happening to an individual based on their current characteristics
- nested cross-validation and grid search.
- logistic regression models

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Predictive modeling in MH disorders

- about high-skew distributional data , longitudinal prospective cohort study
- ADASYN Method and Balanced Down Sampling method
- 5- Fold Cross validation method
- de-anonymization and re-identification, invasive monitoring and untoward influence, and exacerbation of hidden biases and stereotypes
- three types of input data, the BFI-I test, the EMA data, and the Twitter data, using different machine learning models for binary classification.
- SVM model with RBF kernel
- Decision tree models – XGBoost

KEYLEARNINGS

Artificial intelligence in mental health care: a systematic review of diagnosis, monitoring, and intervention applications

- Support Vector Machines
- Random Forest
- The Data Bases - CCTR, CINAHL, PsycINFO, PubMed, and Scopus
- Variable Paradox
- One study found that using more than 15 variables did not significantly improve accuracy, underscoring the critical need for targeted variable selection over complex volume

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Artificial intelligence in mental health care: a systematic review of diagnosis, monitoring, and intervention applications

- Elastic Net Regularization
- An AI robotic puppy (Aibo) used in isolation units significantly decreased cortisol and increased oxytocin
- AI apps using facial recognition significantly improved medication adherence
- Correlation vs. Causation
- Bayesian networks